SIMATS ENGINEERING

ASSESSMENT TOOL 1: Short Answer Test

COURSE NAME: Cloud Computing and Big Data Analytics **COURSE CODE:** CSA15

COURSE OUTCOME COVERED

CO1: *Apply cloud computing technologies including hardware-software evolution, virtualization, web services, and service models to develop cloud-based solutions.* (BL2)

Assessment Tool Weightage: 40%

Dave's Taxonomy: Imitation (Level 1) **Total Marks: 25**

Question Count: 5

Code of Conduct

I **Nevitha M/192521206** (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

Nevitha M

Assessment Tasks

Short Answer Test

1. Define cloud computing and explain any four characteristics that make it different from traditional on-premise computing.
2. Trace the evolution of cloud computing from hardware evolution to internet software evolution, and show how these changes enabled modern cloud platforms.
3. Explain the role of server virtualization in cloud computing. Differentiate between virtualization as an enabling technology and cloud computing as a service model.
4. Discuss the role of **data centers** in delivering cloud services..
5. Differentiate IaaS, PaaS, SaaS, and XaaS with one suitable real-world example for each model.

Short Answer Test Rubric

Criteria	Weightage	Level 4 - Exemplary	Level 3 - Proficient	Level 2 - Developing	Level 1 - Beginning
Conceptual Accuracy	30%	Accurate definitions and precise explanation of all key terms	Mostly accurate with minor gaps	Partial understanding with noticeable errors	Incorrect or irrelevant explanation
Coverage of Concepts	20%	Covers all expected sub-points completely	Covers most expected points	Covers only basic points	Very limited coverage
Use of Examples	15%	Uses highly relevant cloud examples to strengthen the answer	Uses relevant examples with minor mismatch	Uses vague or partially relevant examples	No useful examples
Comparison / Differentiation	20%	Clearly distinguishes concepts with strong logic	Reasonable differentiation with minor overlap	Comparison is weak or incomplete	Fails to differentiate concepts
Clarity of Expression	15%	Clear, organized, and concise answer writing	Generally clear with few issues	Understandable but poorly organized	Difficult to follow

Course code: CSA1515 Name: Nevitha M

Course Name: Cloud Computing and big data Analytics. Reg NO: 192521206

SHORT ANSWER TEST

① Define cloud computing and explain any four characteristics that make it different from traditional on-premise computing.

Definition:-

Cloud Computing is a delivery of Computing Service such as storage, data, access and software over the network (or) Internet.

Four characters of cloud computing:-

- on demand self service - user interface anytime.
- Broad network service - Access through Internet device.
- Resource pooling : Multiple users show the different Computing services.
- Rapid elasticity : Resources can be increased (or) quickly.

Difference :-

Tradition on-premise Computing :-

→ Traditional on-premise Computing requires an organization to purchase and maintain its own. whereas cloud Computing provides resources over the internet.

(2) Trace the evolution cloud computing :-

1. Mainframes - Large Computers served many users
2. personal Computers - Computers are became smaller.
3. Client Server - client accessed servers.
4. virtualization - one server runs multiple VM.
5. Internet - Enabled remote access
6. Modern cloud - Aws and Google cloud.

(3) Server virtualization :-

→ virtualization creates multiple VM on one physical Server

- Improve resource ubilitation
- Reduce hardware costs virtualization

- Creates virtual Machines

Eg: VMware

Cloud computing provides multiple servers.

Eg: AWS.

④ Role of data centers.

- Stores data
- Runs application
- Connects services to the internet
- Protects data
- Maintains service availability.

⑤ IAAS:

↳ Infrastructure as a service: Provide infrastructure.

Eg. Amazon.

PAAS:

↳ platform as a service - provide a build platform application

E.g. Google app.

SAAS - Software as a service - provide ready-to-use software.

E.g. Google

XAAS - anything as a service - provide on IT or any application.